

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and deployed **large-scale data pipelines** using **Python, Apache Spark**, and **AWS Lambda** to process **2TB+ daily transaction data** across payments and banking operations, achieving **99.95% uptime SLA** with automated monitoring via **CloudWatch** and **Splunk**.
- Built **enterprise data warehouse schemas** in **PostgreSQL, MySQL**, and **Oracle** supporting **ISO 20022 message flows** and **SEPA cross-border payment reconciliation**, enabling real-time transaction visibility for **regulatory compliance** and **audit trail requirements** across **GDPR-governed** financial services environments.
- Developed **data quality frameworks** using **pandas, NumPy**, and **scikit-learn** to validate **ETL transformations**, detect **data anomalies**, and perform **statistical profiling** on payment message datasets, reducing data-validation manual effort by **60%** and improving incident detection latency to **5 minutes**.
- Orchestrated **CI/CD data-engineering workflows** via **Jenkins, Git**, and **infrastructure-as-code** practices, managing **code reviews** and **automated testing** for 15+ concurrent **data pipeline deployments**, reducing post-deployment incidents by **40%**.
- Engineered **real-time operational dashboards** in **Power BI** and **Tableau** with **DAX measures** and **data modelling** to surface **SLA metrics, P1/P2 incident trends**, and **payment throughput KPIs** for banking operations teams and senior stakeholders, reducing decision-making latency by **30%**.
- Collaborated with **cross-functional teams** (data analysts, BI engineers, backend services, and change-management) on **data integration projects** across **enterprise business systems** and **REST API-driven microservices**, supporting **UAT coordination** and **post-implementation triage** for mission-critical financial workflows.

SKILLS

Data Engineering & Pipeline Development – Apache Spark, Python, ETL design, data pipeline orchestration, workflow automation, Apache Airflow.

Data Warehousing & SQL – PostgreSQL, MySQL, Oracle, schema design, data modelling, dimensional modelling, query optimization.

Data Quality & Validation – pandas, NumPy, statistical profiling, anomaly detection, data validation frameworks, scikit-learn.

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, CI/CD (Jenkins, Git), infrastructure-as-code, Linux, shell scripting.

Business Intelligence & Analytics – Power BI, Tableau, DAX, Power Query, data visualisation, interactive dashboards, BI modelling.

Financial Services & Regulatory – ISO 20022 message flows, SEPA payments, cross-border payment processing, SWIFT messaging (MT/MX), PSD2, GDPR compliance, audit trails, SLA management.

Monitoring & Incident Management – Splunk, Dynatrace, CloudWatch, production support, P1/P2 incident response, application reliability, observability.

EDUCATION AND TRAINING

PROJECTS

Real-Time Payment Data Warehouse & Analytics Platform (HCL Technologies)

- Architected and deployed a **petabyte-scale data warehouse** leveraging **Apache Spark**, **PostgreSQL**, and **AWS S3** to consolidate **SWIFT MT/MX message streams**, **ISO 20022 transaction flows**, and **SEPA payment records** from 12+ upstream banking systems, enabling unified **cross-border payment analytics** and **regulatory reporting**.
- Built **automated ETL pipelines** in **Python** and **Apache Airflow** processing **500M+ daily payment transactions**, with **data quality checks** using **pandas** and **statistical validation** to ensure **GDPR compliance**, **audit trail integrity**, and **99.99% data accuracy** across all payment message types.
- Designed **real-time operational dashboards** in **Power BI** and **Tableau** with **DAX measures** and **data modelling** surfacing **SLA metrics**, **payment throughput KPIs**, **reconciliation status**, and **ISO 20022 compliance indicators** for operations teams, reducing incident detection time by **45%** and enabling **\$2.5M annual revenue protection** through faster dispute resolution.
- Orchestrated **CI/CD deployment pipelines** via **Jenkins**, **Git**, and **infrastructure-as-code** (Terraform), managing **containerised Lambda-based data processors**, reducing pipeline deployment risk by **35%** and achieving **zero-downtime releases** across production payment workflows.
- Collaborated with **data analysts**, **BI engineers**, **compliance teams**, and **banking domain architects** on end-to-end **data integration**, **schema evolution**, and **regulatory requirement mapping**, ensuring all **PSD2 open-banking** and **SEPA instant payment** standards were reflected in the warehouse schema and analytics layer.

Enterprise Data Quality & Anomaly Detection Framework (HCL Technologies – Banking & Payments Domain)

- Engineered a **machine learning-driven data validation framework** using **scikit-learn**, **statistical analysis** (R / RStudio), and **Python** to automatically detect **data anomalies**, **duplicate payment messages**, and **ISO 20022 schema violations** across **SWIFT payment dashboards**, reducing manual data-quality reviews by **65%**.
- Integrated **Splunk** and **CloudWatch** monitoring alerts with custom **Python ETL validation logic** to trigger **incident response workflows** for **P1/P2 production data quality issues**, achieving **5-minute detection-to-escalation SLA** and preventing **\$1.8M in potential reconciliation errors** over 12 months.
- Deployed **data profiling dashboards** in **Power BI** with **DAX measures** and **Power Query transformations** to visualise data distribution, missing-value trends, and statistical outliers across payment transaction fields, enabling **proactive root-cause analysis** and reducing false-positive incident alerts by **40%**.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)